

EE 311: Device Electronics for Integrated Circuits

Course Code	EE 311
Course Name	Device Electronics for Integrated Circuits
L-T-P-C	3-0-0-3
Prerequisites	IC121 (Mechanics of Particles and Waves)
Intended for	2nd and 3rd year UG
Distribution	Core for 2nd Yr. and 3rd Yr. Electrical Engineering
Semester	Odd
One-time approval	recommended by the 9th BoA (Oct 2015)

Course Contents

- **SEMICONDUCTOR ELECTRONICS** [6 Lectures]
 - i. Physics of Semiconductor Materials
 - ii. Band Model of Solids
 - iii. Carrier distribution functions
 - iv. Free Carriers in Semiconductors, Concept of electrons and holes,
 - v. Concept of equilibrium and non-equilibrium in semiconductor device vi. Current Conduction mechanisms in semiconductors
- **P-N JUNCTIONS** [12 Lectures]
 - i. Fundamentals of p-n junction
 - ii. p-n junction under thermal equilibrium
 - iii. Operation of p-n junction under forward and reverse bias
 - iv. Different type of junctions including step junction, linearly graded junction and heterojunctions,
 - v. Junction Breakdown: Physics of avalanche and Zener breakdown mechanisms vi. Generation and Recombination in a p-n junction
 - vii. Current-Voltage Characteristics of p-n junctions
 - viii. Devices based on p-n junction, Solar cells, LED and photodetectors
- **METAL-SEMICONDUCTOR CONTACTS** [3 Lectures]
 - i. Idealized Metal-Semiconductor junctions
 - ii. Physics of Schottky and Ohmic contacts
 - iii. Effect of surface states on Metal-Semiconductor Contacts, iv. Devices based on metal-semiconductor contacts
- **BIPOLAR TRANSISTORS** [9 Lectures]
 - i. Physics and operation of bipolar junction transistors
 - ii. Current conduction mechanism in bipolar junction transistor
 - iii. Ebers-Moll Model
 - iv. Effects of Collector Bias Variation (Early Effect)
 - v. Small-Signal Transistor Model
 - vi. Operation of bipolar junction transistor under high frequency vii. Devices based on bipolar junction transistor
- **FIELD-EFFECT TRANSISTORS (MOSFETs)** [9 Lectures]
 - i. The ideal MOS Structure
 - ii. Capacitance of the MOS System
 - iii. CV Behavior of a MOS System; Ideal condition, effect of oxide and interface charge
 - iv. Structure and operation of MOSFET devices
 - v. Improved Models for Short-Channel MOSFETs vi. Devices based on MOSFET
- **ELECTRONIC DEVICES AND NANOELECTRONICS** [3 Lectures]
 - i. Electronic Device Materials: Silicon, Germanium, and Gallium Arsenide.
 - ii. Introduction to advanced device technology: Purification and growth, wafer production, epitaxy and deposition, oxidation and metallisation; lithography and implantation
 - iii. Emerging Device Technologies

Text Books

1. S. M. Sze and M.K. Lee, **Semiconductor devices - Physics and Technology**, 3rd Edition, John Wiley & Sons, 2012.

References

1. S. M. Sze and Kwok K.Ng, **Physics of Semiconductor Devices**, 3rd Edition, John Wiley & Sons, 2002.
2. Ben G. Steetman and Sanjay Banerjee, **Solid State Electronic Devices**, 6th Edition, Prentice Hall, 2005
3. Robert F. Pierret, **Semiconductor Device Fundamentals**, Addison-Wesley Publishing, 1996
4. Donald A. Neamen, **Semiconductor Physics and Devices**, 3rd Edition, McGrawHill, 2003
5. Jasprit Singh, **Semiconductor Devices - Basic Principles**, John Wiley and Sons, 2001.